

MOHAMMED AFAAN ANSARI

Curriculum Vitae

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RESEARCH INTERESTS

Large language models (LLMs) and their applications; multimodal AI systems and large vision-language models (LVLMs) for crisis informatics; retrieval-augmented generation (RAG) and multimodal summarization; parameter-efficient fine-tuning (LoRA/PEFT); data visualization intelligence; social media analytics, event detection, and misinformation detection; applied NLP for information retrieval and emergency management.

EDUCATION

Master of Science, Computer Science

Aug 2024 - May 2026

University of Maryland, College Park

College Park, MD

GPA: 3.81 / 4.0

Research Focus: Multimodal LLM optimization, RAG pipeline design, parameter-efficient fine-tuning, and crisis informatics.

Advisor: Prof. Cody Buntain (College of Information Studies)

Bachelor of Technology, Computer Engineering

Jul 2019 - Jun 2022

Sardar Patel Institute of Technology (SPIT), Mumbai University

Mumbai, India

GPA: 8.9 / 10.0

Research Focus: Social network event detection, fake news identification, and blockchain systems.

Capstone research led to peer-reviewed ACM and Springer publications.

Diploma, Information Technology

Jun 2016 - May 2019

Government Polytechnic Mumbai

Mumbai, India

Graduated with distinction; served as undergraduate teaching assistant in programming and IT courses.

PUBLICATIONS

Conference Papers & Book Chapters

[1] **Mohammed Afaan Ansari**, Aniruddh Bansal, Tianyi Zhou. “ChartDesign: Towards LLM Designer of Data Visualization.” *Under review, Conference on Language Modeling (COLM) 2026*.

[2] **Mohammed Afaan Ansari**, Jiten Oswal, Vivek Kulkarni, Ashiqur R. KhudaBukhsh. “Quantifying the Transience of Social Web Datasets.” *Proceedings of the 2023 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2023)*. **Awarded 2nd Best Paper.** [\[ACM DL\]](#)

[3] Pramod Bide, Sudhir Dhage, **Mohammed Afaan Ansari**, Rudresh Veerkhare. “Cross Event Detection and Topic Evolution Mining in Cross Events for Man Made Disasters in Social Media Streams.”

[4] Pramod Bide, Sudhir Dhage, **Mohammed Afaan Ansari**, et al. “Fake News and Fake Profile Detection Using Hybrid Machine Learning on Social Media.”

[5] **Mohammed Afaan Ansari**, et al. “Securing Blood Distribution Systems Using Blockchain.” *Springer Smart Innovation, Systems and Technologies (SIST), International Conference on Computing and Emerging Technologies (ICCET 2021)*. [\[Springer\]](#)

Manuscripts in Preparation

[6] **Mohammed Afaan Ansari**. “ARIA: Autonomous Risk Intelligence Agent for Vulnerability Prioritization.” *Technical report / system design, 2026*. A 10-agent agentic AI system for business-risk-aware CVE prioritization; integrates NVD, EPSS, CISA KEV, MITRE ATT&CK, and compliance frameworks (PCI DSS, HIPAA, SOC2) to translate technical vulnerability data into per-CVE ROI calculations and remediation plans with full audit trails.

[7] **Mohammed Afaan Ansari**. “Efficient Vision-Language Model Inference for Image-Grounded Gen-

RESEARCH EXPERIENCE

Research Assistant - Multimodal LLMs & Crisis Informatics

Jan 2025 - Present

University of Maryland, College Park - CLIP / iSchool

College Park, MD

Advisor: Prof. Cody Buntain, College of Information Studies

Project: *Automatically Generating Traceable, Multimodal Summaries of Crises and Conflict*

- **LVLM benchmarking for crisis fact extraction:** Systematically evaluated 9 large vision-language model families - InternVL 2.5 MPO (8B), Google Gemma 3 (27B), DeepSeek Janus Pro (7B), Phi 3.5 Vision-Instruct (4.2B), Llava 1.6 Mistral/Vicuna, Llava 1.5, Llama 3.2 Vision (11B), Mistral Pixtral (12B) - on image-based crisis fact extraction using the CrisisFACTS and Incident Streams datasets. InternVL 2.5 produced the highest proportion of high-importance facts (~50%) while maintaining competitive runtime (28 s/image vs. 295 s for Gemma 3).
- **Multimodal FACTS pipeline:** Built and packaged the full end-to-end pipeline: multimodal content ingestion (images + text) → LVLM-based fact extraction with event-type-specific query prompts → Mistral-7B-Instruct summarization → embedding-based deduplication (SentenceTransformer, cosine > 0.9) → JSON fact-list output in CrisisFACTS submission format. Produced submissions for events covering the Beirut explosion, Houston explosion, and Hurricane Dorian.
- **Priority classification:** Trained a RoBERTa-large binary classifier on 126,360 priority-annotated Incident Streams Twitter posts (macro-F1 = 0.77; F1 = 0.95 low-priority, 0.60 high-priority) to automatically assess the importance of image-extracted facts and enable scalable ranking of LVLM outputs without human review.
- **LLM-based fact assessment:** Benchmarked 6 LLM assessors - Mistral 7B, Llama 3 7B, Llama 3.1 80B (quantized), Ministral 8B, Mistral 7B v0.3, Qwen 3. Ministral 8B achieved average Pearson $r = 0.346$ while fitting in ~20 GB VRAM vs. ~100 GB for Llama 3.1 80B; LLM ranking stability confirmed at $\rho = 0.79$ (Spearman), and Pearson $r = 0.83$ for F1 scores.
- **Text pipeline & comparison:** Implemented and evaluated a text-only fact-extraction pipeline using PyTerrier-based query relevance scoring and InternVL-2.5-chat as the text processing module; compared pipeline variants across multiple scoring dimensions including priority scoring, cosine-similarity reranking, and source-bonus weighting.

Research Assistant - Multimodal Summarization (ChartDesign)

Aug 2024 - Present

University of Maryland, College Park

College Park, MD

- Developed **ChartDesign**, a framework post-training compact LLMs (Phi-3 mini 4B, Qwen-3 8B, InternVL2.5 8B) with LoRA adapters to generate renderer-agnostic JSON design specifications from tabular data, covering chart type, orientation, axis labels, spacing, and visual style attributes.
- Curated a 2,118-chart training corpus (PewResearch + CharXiV) with paired CSV data and structured design-attribute JSONs; devised inverse-frequency sampling strategy (importance weight $w_k = \log(1 + N/n_k)$) to address class imbalance across 15+ design attributes.
- Achieved 84% overall attribute accuracy on held-out test set (vs. 55% for best zero-shot baseline); validated LLM judge at 92% agreement with human raters; 65% of annotators preferred ChartDesign-generated charts over human originals.

Research Assistant - Social Data Mining

Jul 2022 - Jun 2023

Rochester Institute of Technology

Rochester, NY (Remote)

Advisor: Prof. Ashiqur R. KhudaBukhsh, Golisano College of Computing and Information Sciences

- Co-authored ACM paper measuring lifespan and dataset drift in social web corpora using BERT-based models on longitudinal Twitter and Reddit data snapshots; work awarded **2nd Best Paper** at ASONAM 2023.
- Built automated data-collection pipelines (Twitter API, Reddit Pushshift) capturing longitudinal snapshots across 100+ crisis and social events to study content transience.

Research Assistant - Machine Learning & LLM Systems

Dec 2022 - Dec 2023

Georgia Institute of Technology & Mayo Clinic

Atlanta, GA (Remote)

Advisor: Prof. Joy Arulraj, School of Computer Science, Georgia Tech

- Developed CNN-based MRI brain tumor detection model achieving 85% accuracy across 1,000 patients;

applied augmentation and class-weighted training to handle severe label imbalance; system expected to be integrated into Mayo Clinic's standard clinical practice.

- Benchmarked GPT-4 API vs. ONNX-quantized vision encoders within [EVADB](#) - an open-source visual AI database enabling low-code querying of ML models - reducing per-image inference latency by 40% (150 ms → 90 ms) and enabling sub-second clinical queries. Also developed a text-document querying pipeline integrated into EVADB's HuggingFace model hub.
- Implemented OpenAI-LLM and object-detection frameworks within EVADB, providing a competitive low-code alternative to LangChain and PandasAI for multimodal data querying.

Undergraduate Research Intern - Machine Learning & NLP

Jan 2020 - Jun 2022

Sardar Patel Institute of Technology (SPIT)

Mumbai, India

Advisor: Prof. Pramod Bide & Prof. Sudhir Dhage, Department of Computer Engineering

- Designed the Cross Event Evolution Detection (CEED) framework: tweet segmentation via Wikipedia titles DB, Jarvis-Patrick clustering, Wikipedia-backed event-worthiness scoring, and temporal cross-event detection; validated on 278,817 tweets from the 2019 Pulwama attack news cycle.
- Developed the Topic Evolution Algorithm (TEA) tracking topic fulcrum shifts across 10 sub-windows per event using inverse-frequency-weighted TF-IDF similarity.
- Built a hybrid ML fake news and fake profile detection system combining LSTM language models with network-topology features; work led to Springer publication.
- Co-authored a blockchain-based secure blood distribution system paper, published at ICCET 2021 (Springer SIST).

INDUSTRY & APPLIED EXPERIENCE

Data Science Intern

Jun 2025 - Aug 2025

CSSI

Washington, D.C.

- Engineered anomaly-detection models on 10M+ FAA aviation-weather records (90%+ recall), enabling analysts to identify hazardous flight patterns 2× faster.
- Built FAA-compliant ETL pipelines and REST API integrations for cross-system compliance dashboards; presented findings to senior FAA stakeholders.

Research Participant - Green CapEx Initiative

Jul 2025 - Sep 2025

Morgan Stanley Capital International (MSCI) Sustainability Institute

Remote

- Generated ESG and CapEx metrics from 10,000+ financial records supporting MSCI's Green CapEx taxonomy, adopted in sustainability research and institutional investor reporting.

Data Science Technology Associate

Dec 2023 - Aug 2024

Morgan Stanley Capital International (MSCI Inc.)

Mumbai, India

- Promoted to Associate in 1 year vs. the typical 3-4 year track; recognized as **Star Performer 2023** among 1,200+ employees across all levels.
- Led development of GCP-based extraction pipeline using GCP AI models to extract data from clients' financial reports, establishing MSCI's first hybrid cloud system; set up an additional correction flow for incorrectly classified data.
- Achieved 90%+ extraction accuracy from financial records, significantly surpassing the existing manual baseline of 60%; analyzed insights to inform resource allocation, dataset management, and model deployment decisions.
- Devised prompt engineering strategies for Turnover/Revenue use cases to replace model training, reducing cost from \$10 per 1,000 items to \$0.002 - a 99.98% reduction - via ONNX-optimized inference (<100 ms latency).
- Managed resource allocation analysis across datasets and models, identifying bottlenecks and overcoming data variation challenges at scale (100,000+ research reports).

Data Science Technology Analyst

Jul 2022 - Dec 2023

Morgan Stanley Capital International (MSCI Inc.)

Mumbai, India

- Built end-to-end GCP data pipeline for processing 100,000+ research reports, enabling MSCI's first hybrid cloud deployment and reducing QA errors by 30%.

- Applied Gemini-based AI models for document intelligence and extraction tasks; collaborated cross-functionally with data engineering, product, and compliance teams.
- Designed monitoring and alerting infrastructure for pipeline health and model performance tracking across production environments.

Lead Software Developer

Sourceable Inc. (Columbia University-backed startup)

Jan 2022 - Jul 2024
New York, NY (Remote)

- Led front-end and full-stack development of a real-time crisis-mapping platform (ReactJS, React Native, Node.js, Docker, Kubernetes, Dialogflow) adopted by 1,500+ journalists across 52+ countries for real-time crisis reporting in humanitarian emergencies.
- Reduced page load times by 30% through component optimization and lazy loading; improved user retention by 25% via UX-driven feature iterations informed by journalist feedback.
- Managed CI/CD pipelines ensuring zero-downtime releases; coordinated cross-functional team of engineers, designers, and content specialists across multiple time zones.

Technology Analyst Intern

Nomura Holdings - Global Markets IT

Jan 2022 - Jun 2022
Mumbai, India

- Designed and implemented custom monitoring rules using the ELK stack (Elasticsearch, Logstash, Kibana) for detecting ongoing traded-data irregularities across equities trading infrastructure; reduced incident resolution time 40%.
- Built and deployed an AI-powered smart mirror system for personalized financial instrument messages and daily market updates; system deployed across Nomura's India offices.
- Placement via Kakushin Hackathon (Runner-Up); received full-time return offer with 40% salary increase compared to the incoming analyst batch.

Angular & Java Developer Intern

Sven Consultants & Technology Pvt. Ltd.

Jun 2021 - Aug 2021
Mumbai, India

- Developed Angular 8 web applications and Java backend services; participated in enterprise software development lifecycle including requirements gathering, coding, and testing.

App Developer (Contract)

ECS Diploma & Engineering Classes

Mar 2020 - Jun 2020
Mumbai, India (Remote)

- Built a live lecture and content-delivery application during COVID-19 lockdown, enabling remote learning for hundreds of students across diploma and engineering programs.

Web Developer

Pen Mouse Design Technology Pvt. Ltd.

Jun 2019 - Aug 2019
Mumbai, India

- Developed and maintained company web presence; implemented responsive UI components and integrated content management workflows.

TEACHING & ADVISING

Teaching Assistant - Undergraduate Courses

Teaching Assistant, Data Structures & Algorithms

Sardar Patel Institute of Technology (SPIT), Mumbai University

Jan 2021 - Jun 2022
Mumbai, India

- Conducted weekly lab sessions (40-60 students); developed supplementary problem sets; graded assessments; received student feedback rating of 4.6/5.

Teaching Assistant, Introduction to Machine Learning

Sardar Patel Institute of Technology (SPIT), Mumbai University

Aug 2021 - Dec 2021
Mumbai, India

- Facilitated Python programming labs (scikit-learn, NumPy, neural networks) for 35+ students under Prof. Pramod Bide; designed project assignments on text classification and regression.

Teaching Assistant, Programming Fundamentals (C/C++ & Web Dev) Jun 2018 - May 2019
Government Polytechnic Mumbai Mumbai, India

- Supported instructors in teaching introductory programming to Diploma IT cohort (30+ students); conducted lab sessions and prepared tutorials on C pointer arithmetic and basic data structures.

Mentorship & Workshops

Student Mentor - ML / Data Science 2021 - Present

Mentored 40+ undergraduate and early-career ML/DS students on project design, interview preparation, and research methodology; 10+ placed at FAANG or high-growth startups.

Workshop Facilitator - “Large Language Models: From Theory to Deployment” Apr 2024
Virtual (60+ participants) - Designed and delivered a 3-hour hands-on workshop covering transformer architecture, LoRA fine-tuning, and RAG pipeline construction using LangChain.

Workshop Facilitator - “Social Media & NLP: Research Methods” Nov 2022
SPIT Technical Symposium, Mumbai - 2-hour session for 50+ students on Twitter API, text preprocessing, and topic modelling (LDA, BERT); distributed open-source code notebooks.

Volunteer, DTE FDP Programme on Big Data & Hadoop Cluster Administration
Directorate of Technical Education (DTE), Maharashtra - Volunteered at faculty development programme on big data infrastructure and Hadoop cluster administration.

SELECTED PROJECTS

ChartDesign: LLM-Powered Data Visualization Designer 2024 - Present
Post-trained Phi-3, Qwen-3, and InternVL2.5 via LoRA to map CSV tables to renderer-agnostic JSON design specifications. 84% attribute accuracy; renderings work in Matplotlib, Vega-Lite, Altair, ggplot2. (*Under review, COLM 2026.*)

Multimodal FACTS Crisis Summarization Framework 2025 - Present
End-to-end pipeline: LVLm fact extraction (9 models benchmarked) → RoBERTa priority scoring → Ministral 8B LLM assessment → JSON submission output. Evaluated image-based fact utility against text-only baselines across multiple crisis event scenarios.

CEED: Cross Event Evolution Detection 2020 - 2022
Real-time social event-mining on Apache Flume + HDFS; bursty-segment extraction, Jarvis-Patrick clustering, cross-event similarity and Topic Evolution Algorithm. Validated on 278K tweets.

Blockchain-Based Secure Blood Distribution System 2021
Designed a Hyperledger Fabric blockchain solution for end-to-end blood bank supply-chain integrity; verified donation, storage, and recipient authenticity on a permissioned ledger. (*Published, Springer ICCET 2021.*)

UNet Diffusion Model for Low-Light Image Restoration 2024
UNet-conditioned DDPM for low-light enhancement on a 3,000-image smartphone dataset; PSNR +3 dB, SSIM +0.02 vs. baseline; mixed-precision ONNX inference meeting <150 ms mobile GPU latency.

ARIA: Autonomous Risk Intelligence Agent 2026
Designed a 10-agent agentic AI system for business-risk-aware CVE vulnerability prioritization. Agents autonomously ingest NVD/EPSS/CISA KEV data, map CVEs to organizational assets via natural-language business context documents, estimate compliance fine exposure (PCI DSS, HIPAA, SOC2), simulate adversarial blast-radius paths through service dependency graphs, and calculate per-CVE ROI. Built on Claude Sonnet + FastAPI + React; validated against HHS Breach Portal. Projected 70-80% reduction in manual triage labor vs. industry-standard CVSS-only approach.

LLM QnA System for Enterprise Knowledge Bases 2023
Domain-adaptive QnA engine (GPT-4 + BM25 + reranking) for enterprise knowledge bases; designed retrieval pipeline combining dense and sparse retrieval with cross-encoder reranking for accurate, citation-backed responses over large document corpora.

TALKS, CONFERENCES & SERVICE

Conference Attendance & Presentations

TechGiving Conference

2024

Washington, D.C. - Attended technology-for-social-good conference focused on AI applications in humanitarian and civic contexts.

ASONAM 2023 (ACM/IEEE)

Nov 2023

Marrakech, Morocco - Presented “Quantifying the Transience of Social Web Datasets”; **Awarded 2nd Best Paper.**

Academic Service

Hackathon Judge / Evaluator

2021 - Present

- SPIT Hackathon (2021, 2022, 2023) - Technical judge, AI/ML and blockchain tracks.
- Codeshastra, DJ Sanghvi (2021, 2022) - Judge for data analytics and Web3 tracks.
- Smart India Hackathon (SIH) - Internal evaluator at SPIT department round.

Community & Volunteer Service

NGO Technology Advisor & Fundraising Lead

2021 - 2023

Designed digital outreach and donor CRM workflows for a community-development NGO; increased annual donations by 70% (\$50k incremental), funding 3 new community programs.

Social Work at Computer Centre, SPIT

2019 - 2021

Completed 50+ hours of voluntary social service at the college computer centre, assisting students with digital literacy and access.

AWARDS & HONORS

Research Awards

- **2nd Best Paper Award**, ASONAM 2023 - “Quantifying the Transience of Social Web Datasets” (Nov 2023)
- **Certificate of Excellence**, SPIT - For winning technical event at IIT Bombay (national competition)
- **Certificate of Excellence**, SPIT - For winning technical event at Nomura Holdings
- **Certificate of Excellence**, SPIT - For winning technical event at KJ Somaiya

Hackathon & Competition Wins

- **Winner**, SPIT Hackathon - Payroll management system (2021)
- **Winner**, Codeshastra Hackathon, DJ Sanghvi - Blockchain track (2021)
- **Winner**, App Development Competition, FSAI (2021)
- **Winner**, State-Level Technical Paper Presentation (TPP) - TECH KNOW
- **Winner**, State-Level Technical Paper Presentation (TPP) - PRAVINYA
- **Pharma Hackathon Winner** (2022)

Runner-Up / Finalist Positions

- **Runner-Up**, Kakushin Hackathon, Nomura Holdings - Fitness track (placement led to Nomura internship)
- **2nd Runner-Up**, KJSCE Hack 6.0, KJ Somaiya - Blockchain track
- **2nd Runner-Up**, TPP DYP-Techno (state-level technical paper presentation)
- **2nd Runner-Up**, TPP TECHNEX (state-level technical paper presentation)
- **Finalist**, Hack-Innovate Global International Hackathon, Nomura Holdings (employee/invited competition)
- **Semi-Finalist**, IIT Bombay Eureka! - Entrepreneurship & Innovation programme
- **Semi-Finalist**, Smart India Hackathon (SIH), Government of India
- **Finalist**, Code Break Hackathon, MIT Pune
- **Finalist**, TSEC Hackathon, Thadomal Shahani Engineering College
- **Finalist**, Hack Infinity, DAIICT
- **Finalist**, ETH India Hackathon (Ethereum India)
- **Finalist**, Hack Chennai Hackathon
- **Finalist**, Hackon Hackathon

TECHNICAL SKILLS

Generative AI & LLMs: PyTorch, HuggingFace Transformers, LoRA/PEFT, RAG pipelines (LangChain, LlamaIndex), ONNX, TorchScript, vLLM; fine-tuned multimodal LLMs (Phi-3, Qwen-3, InternVL, LLaVA); mixed-precision inference, quantization, GPU profiling.

Computer Vision & Multimodal: OpenCV, CLIP, diffusion models (UNet, DDPM), GANs; image restoration (denoising, super-resolution); PSNR/SSIM evaluation; mobile GPU deployment.

NLP & Data: BERT, RoBERTa, GPT family, T5; text classification, NER, topic modelling (LDA, BERTopic); BERTScore, entailment scoring (DeBERTa); social media APIs (Twitter, Reddit); pandas, NumPy, scikit-learn, PyTerrier, Matplotlib, Seaborn.

MLOps & Cloud: GCP (BigQuery, Vertex AI), AWS (SageMaker, S3), Docker, Kubernetes, Airflow, CI/CD (GitHub Actions), REST APIs, Terraform, ELK stack.

Web & Software Engineering: Python, JavaScript/TypeScript, React, Angular, Node.js, SQL; Blockchain (Hyperledger Fabric, Ethereum/Solidity); Agile/Scrum; Figma.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Deep Learning & Machine Learning Specializations

- *Neural Networks and Deep Learning* - Coursera / DeepLearning.AI (Andrew Ng)
- *Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization* - Coursera / DeepLearning.AI
- *Structuring Machine Learning Projects* - Coursera / DeepLearning.AI
- *Convolutional Neural Networks* - Coursera / DeepLearning.AI
- *Sequence Models* - Coursera / DeepLearning.AI
- *Introduction to Data Science in Python* - Coursera / University of Michigan
- *Applied Plotting, Charting & Data Representation in Python* - University of Michigan
- *Applied Machine Learning in Python* - University of Michigan
- *Applied Text Mining in Python* - University of Michigan
- *Applied Social Network Analysis in Python* - University of Michigan
- *Classification with Transfer Learning in Keras* - Coursera

Cloud, Systems & Blockchain

- *End-to-End Machine Learning with TensorFlow on Google Cloud* - Advanced ML on GCP Specialization, Coursera
- *Production Machine Learning Systems* - Advanced ML on GCP Specialization
- *Computer Vision Fundamentals with Google Cloud* - Advanced ML on GCP Specialization
- *Natural Language Processing on Google Cloud* - Advanced ML on GCP Specialization
- *Recommendation Systems on Google Cloud* - Advanced ML on GCP Specialization
- *Blockchain Development on Hyperledger Fabric using Composer* - Udemy / Linux Foundation
- *Cloud Computing* - NPTEL, IIT Kharagpur
- *C Programming - Spoken Tutorial* - IIT Bombay
- *Advanced Data Structures and Problem Solving using Java* - Suven Consultants & Technology
- *Angular 8 Development* - Suven Consultants & Technology
- *JPMorgan Chase Software Engineering Virtual Experience* - Forage
- *Ethical Hacking* - ACIT; IETE-TSEC; Kyron Technologies, IIT Bombay (three separate workshops)